LEVERAGING AI TOOLS IN RF AND SRF OPERATIONS AT FRIB*

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Abstract

Radio-frequency (RF) and superconducting radio-frequency (SRF) operations at the Facility for Rare Isotope Beams (FRIB) require rapid, high-consequence decision-making across low-level RF controls, cryomodules, tuners, interlocks, diagnostics, cryogenics, and machine-protection interfaces. In many troubleshooting scenarios the bottleneck is not diagnostic-signal availability, but is the time required to locate relevant context, reconstruct event timelines, interpret heterogeneous data, and transfer institutional knowledge across shifts and teams. This paper proposes a responsible generative artificial intelligence (AI) framework for RF/SRF operations that combines retrieval-augmented generation, multimodal interpretation, and structured workflow acceleration. Three use cases are emphasized: faster RF trip triage, high-efficiency raw-data collection and visualization, and documentation support. A four-level architecture is introduced for an RF/SRF operational brain: event and raw-data acquisition, data cleaning and visualization, document-grounded assistance, and expert-reviewed knowledge generation. AI outputs are treated as draft decision-support artifacts that must be verified by personnel, measurements, and approved procedures.

INTRODUCTION

Modern accelerator facilities operate at the intersection of RF engineering, superconducting cavity physics, cryogenics, controls, beam dynamics, and machine protection. FRIB's driver linac contains 324 superconducting resonators housed in 46 cryomodules and supports heavy-ion beam delivery above 200 MeV/u for rare-isotope production [1, 2]. RF/SRF operation therefore requires operators and system experts to reason over live process variables, low-level RF (LLRF) waveforms, alarm histories, electronic logbook entries, maintenance records, procedures, and hardware configuration details.

The central operational challenge addressed here is information latency. During an RF trip, field-emission study, tuner problem, or cavity turn-on, relevant evidence may exist but be distributed across archived process variables, waveform files, screenshots, shift summaries, and institutional knowledge. The time required to assemble this context can dominate the time required to make a technically sound decision. Generative AI provides a practical mechanism to reduce this latency, provided that the system is grounded in facility data and procedures, bounded by safety constraints, and used as decision support rather than as an authority.

Recent work in accelerator and scientific-facility AI motivates this direction. Deep-learning methods have been used to classify SRF cavity faults from RF waveform captures at Jefferson Lab [3]. Machine-learning optimization has been used to reduce field-emission-induced radiation while preserving required linac energy gain [4]. Retrieval- and tool-augmented language-model systems have been proposed for scientific facilities [5], and language-model-driven accelerator assistants have demonstrated archive retrieval, process-variable resolution, script generation, and auditable workflow execution in accelerator environments [6, 7]. This paper adapts these ideas to the specific needs of RF and SRF operation at FRIB.

OPERATIONAL MOTIVATION

FRIB's SRF linac cavities include quarter-wave resonators (QWRs) and half-wave resonators (HWRs) operated under different cryogenic and RF conditions. The QWR cryomodules in the first linac segment are operated at 4 K, while HWR cryomodules downstream are operated at 2 K [2]. Stable cavity amplitude and phase control have been demonstrated with slow tuner systems, but practical operation remains constrained by field emission, multipacting, tuner limits, cryogenic stability, interlock thresholds, and evolving beam-power requirements [1, 2]. Figure 1 shows the layout of the FRIB driver linac.

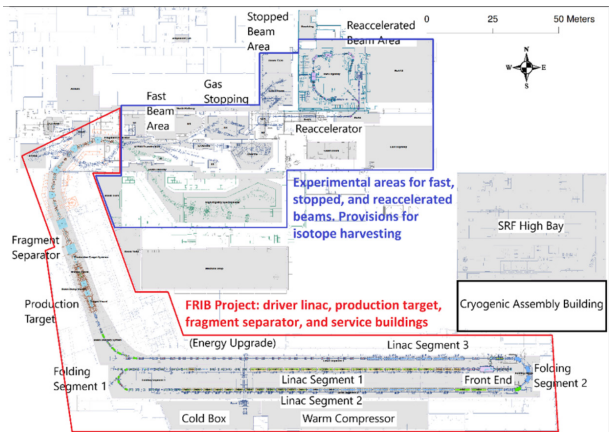


Figure 1: Layout of the FRIB driver linac, target, fragment separator and experimental areas.

Field emission is a representative example of an operationally important SRF problem. RF pulse conditioning has been used at FRIB to reduce field emission in cryomodules, with recent work reporting conditioning and analysis for roughly 50 cavities and distinguishing rapid electrical breakdown events from slower thermal breakdown or quench-like events [8]. Such events generate large volumes of waveforms, X-ray trends, cavity-gradient information,

* Work supported by the U.S. Department of Energy, Office of Science under Cooperative Agreement DE-SC0023633, the State of Michigan, and Michigan State University.
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operator notes, and post-analysis waveform plots. A multimodal generative AI assistant can help organize these data into structured summaries, compare new events with historical behavior, and prepare draft reports for expert review.

RF trip triage provides a second strong motivation. At CEBAF, digital LLRF waveform captures from C100 cryomodules were used to train deep recurrent and convolutional neural networks for cavity identification and fault classification [3]. That work demonstrates the value of automated waveform interpretation, but an operational response also requires context: which interlock occurred first, whether vacuum or coupler signals changed, whether similar events occurred recently, which procedure applies, and what mitigations were previously successful. Generative AI can connect numerical classification to human-readable, evidence-based operational context.

PROPOSED RF/SRF AI WORKFLOW

The proposed framework is a human-supervised decision-support layer, which is not autonomous. It is organized into four levels that separate raw evidence, processed data, document-grounded reasoning, and expert-approved knowledge generation.

Level 1: Event and Raw-Data Acquisition

The first level captures original operational evidence with provenance. Sources include EPICS process variables, archived RF and cryogenic trends, LLRF waveform plots, interlock states, alarm snapshots, electronic logbook entries, screenshots, and manually entered shift notes. Each record retains timestamp, source, cavity or cryomodule association, operating mode, beam state, and configuration metadata. This level preserves evidence rather than interpret it.

Level 2: Data Cleaning, Processing, and Visualization

The second level converts raw evidence into engineering-ready views. For RF trip analysis this may include waveform alignment, pre-trip and post-trip extraction, cavity field decay estimates, forward and reflected power comparison, tuner-pressure overlays, and interlock timing reconstruction. For field-emission analysis it may include X-ray trend extraction, gradient-scan comparison, event labelling, and conditioning-effect summaries. The output includes plots, tabular summaries, and machine-readable files suitable for later review.

Level 3: Document-Grounded Assistance

The third level applies retrieval-augmented generation (RAG) to approved procedures, RF/SRF documentation, LLRF controller notes, maintenance records, commissioning reports, and troubleshooting guides. RAG is well suited to facility operations because it retrieves relevant document passages and uses them as context for answer generation, reducing reliance on model-internal memory and allowing the knowledge base to be updated as procedures evolve [9, 10]. For accelerator operations, every response distinguishes measured evidence, retrieved procedural guidance, and AI-generated interpretation.

Level 4: Knowledge Generation

The fourth level converts verified outcomes into reusable knowledge. After an event is resolved, the assistant can generate a draft event summary, causal chain, corrective actions, relevant plots, affected hardware, and suggested tags. RF personnel review and approve the result before it becomes part of the RF/SRF knowledge base. Over time, this process captures institutional knowledge in a form that is searchable, auditable, and useful for shift handoffs, training, and future troubleshooting.

USE CASES

RF Trip Triage

A trip-triage assistant would automatically gather event-adjacent data and produce a draft package containing trip time, affected cavity or station, first interlock, waveform thumbnails, tuner state, vacuum response, recent alarm sequence, related logbook entries, and similar historical events. The assistant does not declare a root cause without evidence. Instead, it ranks hypotheses such as cavity trip, electrical/thermal breakdown event, field-emission activity, microphonics-induced control saturation, coupler or vacuum activity, or LLRF configuration error. The ranking is accompanied by the observations that support or weaken each hypothesis.

Raw-Data Collection and Visualization

RF/SRF studies often require repeated retrieval of archiver data, waveform files, and logbook context. A controlled AI workflow can generate reproducible analysis scripts, quick-look plots, and standardized data products from natural-language requests. The accelerator-assistant model demonstrated at a production synchrotron is relevant because it combines natural-language task definition, process-variable resolution, archive retrieval, generated scripts, and structured artifacts under bounded execution [6]. For FRIB, similar workflows could support cavity turn-on studies, tuner characterization, field-emission scans, conditioning events, and LLRF performance trending.

Documentation and Shift Support

A document-grounded assistant can answer operational questions using approved RF/SRF procedures and historical records. Example questions include: "*Which procedure applies after this interlock?" "Which cavities have shown similar X-ray behavior?" "What was changed during the last maintenance period for this cryomodule?" "What should be included in the shift handoff?"* The assistant can also draft shift summaries and issue-tracker entries, reducing administrative burden while improving consistency. All outputs remain editable drafts and include citations or links to the underlying records.

RESPONSIBLE DEPLOYMENT

Responsible deployment is essential because RF/SRF operations involve hardware protection, personnel safety, beam availability and delivery to scientific users. Initial deployments are read-only and focused on retrieval, summarization, visualization, and report drafting. Any future tool execution must be limited by authentication, role-based access, process-variable whitelists and blacklists, bounded operating ranges, approval gates, and audit logs. Plan-first orchestration, structured artifacts, and operator approval are important design patterns demonstrated in recent accelerator AI systems [6].

A practical governance model requires that AI-generated conclusions be labelled as draft decision-support content. The assistant states uncertainty, identify missing evidence, and avoids recommending actions that are not supported by approved procedures. Engineering judgment remains authoritative. The most useful AI system for RF/SRF operations is therefore not a general chatbot, but a constrained assistant that accelerates evidence gathering, produces reproducible artifacts, and helps experts communicate decisions.

EXPECTED RESULTS AND IMPLICATIONS

The expected near-term benefits are reduced time to locate procedures, faster reconstruction of RF trip timelines, more consistent shift summaries, and less repeated manual data extraction. Medium-term benefits include improved continuity between shifts, better onboarding of new RF/SRF personnel, improved reuse of historical troubleshooting experience, and more systematic capture of cavity and cryomodule operating history. Future FRIB upgrades and higher-gradient SRF systems will increase the value of such structured knowledge capture because operating margins, field-emission behavior, resonance control, and documentation complexity will continue to grow [11].

The proposed framework contributes by joining two complementary AI directions: numerical machine learning for RF waveform and cavity-behavior analysis, and generative AI for document-grounded reasoning, summarization, and workflow acceleration. In RF/SRF operations, the highest value comes from integrating these capabilities around human decision-making. The system helps experts see the right data faster, remember prior experience accurately, and communicate operational status clearly.

CONCLUSION

Generative AI can streamline RF and SRF operations at FRIB by reducing the time required to find, interpret, and communicate operational context. A four-level RF/SRF operational brain is proposed: event and raw-data acquisition, data processing and visualization, document-grounded assistance, and expert-reviewed knowledge generation. The approach is intentionally human-centered. AI accelerates evidence gathering and draft generation, while measurements, approved procedures, machine-protection systems, and RF/SRF personnel remain authoritative. As accelerator facilities grow in complexity, document-grounded and multimodal AI assistants can become valuable operational infrastructure for reliable, transparent, and knowledge-preserving RF/SRF operations.

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